

Key Ideas A3

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Multilingual Data Balancing: To prevent the model from effectively ignoring the much smaller Hindi and Kannada datasets during training (which is dominated by the massive English corpus), I implemented an oversampling strategy. Indic training examples were repeated $5\times$ per epoch. This ensured the model dedicated sufficient updates to learning cross-lingual alignment and language-specific features.

Q1:

- **Explicit Boundary Representations:** I inserted four special boundary tokens ($[E1]$, $[/E1]$, $[E2]$, $[/E2]$) around the entities in the text. I then extracted the last-layer hidden states at all four of these marker positions and concatenated them into a single comprehensive vector (4×1536). This explicitly guides the classifier head with high-quality representations of both entities that have dynamically attended to the full context, without needing span extraction.
- **Class Imbalance Handling:** To tackle the highly skewed distribution of relation classes, I applied inverse-frequency weights to the cross-entropy loss function.
- **Differential Learning Rates:** Since the classification head was randomly initialized while the base model was pre-trained, the head requires faster adaptation. I assigned the classifier head a $5\times$ higher learning rate compared to the LoRA weights, using a cosine schedule for the backbone, allowing the head to quickly learn to utilize the emerging representations.

Q2:

- I framed the task to generate *only* the final relation label. Using chat-template prompts, the loss was heavily masked over the input and computed solely on the target label tokens. This kept the output space extremely small and prevented the model from wasting capacity learning JSON structures or re-generating inputs.
- Rather than just applying LoRA to the query (q) and value (v) projections, I applied it across all four attention projections (q , k , v , o).

Q3:

- The single most impactful choice was utilizing the BAAI/bge-m3 model for retrieval embeddings. Because it produces highly aligned cross-lingual embeddings, a query in a language with no training data (like Tulu or Oriya) could successfully retrieve genuinely relevant English and Indic demonstrations. I built a FAISS index over the combined demonstration pool to rapidly retrieve the nearest neighbors for every query.